Graph Positional Encoding and Variance-Weighted Learning as Universal Plug-ins for Uplift Modeling

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Abstract

Estimating Individual Treatment Effects (ITE) from networked observational data is crucial for personalized recommendation, yet existing graph-based methods are monolithic—improvements to one architecture do not transfer to others. We introduce two orthogonal, plug-and-play modules that enhance *any* graph ITE estimator: (1) Graph Positional Encoding (GPE), a multi-head self-attention fusion of node features and node2vec positional embeddings that captures community-level structural information, and (2) Variance-Weighted Learning, an inverse-variance sample weighting scheme that mitigates heteroscedastic noise, with both a learned variant for continuous outcomes and an analytical Bernoulli variant for binary outcomes. We conduct a controlled "plug-in" evaluation on eight base models—from the foundational NetDeconf (WSDM 2020) to the current state-of-the-art GDC (WSDM 2025)—across five graph datasets and six real-world tabular uplift benchmarks, totaling over 1,100 experimental runs. GPE improves PEHE in 69% of all graph settings, with TARNet+GPE achieving a perfect 11/11 win rate. On continuous real-world outcomes, variance weighting yields a 4× Qini improvement. At high noise, the combination of both modules reduces GDC's PEHE by up to 31%. Our results establish these modules as a general-purpose enhancement paradigm for graph-based uplift modeling.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Neural networks*; • **Mathematics of computing** → *Probabilistic inference*.

Keywords

uplift modeling, individual treatment effect, graph neural networks, positional encoding, heteroscedasticity, causal inference

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1 Introduction

Accurate estimation of ITE and Conditional Average Treatment Effect (CATE) is fundamental for data-driven decision-making in domains from personalized marketing [23] to promotional campaign optimization [12]. Traditional meta-learners such as S-learner [19] and Doubly Robust (DR) learner [18] typically assume independent and identically distributed data. However, this assumption ignores the complex interdependencies inherent in real-world networked systems, where user behaviors are shaped by both individual traits and peer influence [11, 31]. As illustrated in Figure 1, users within

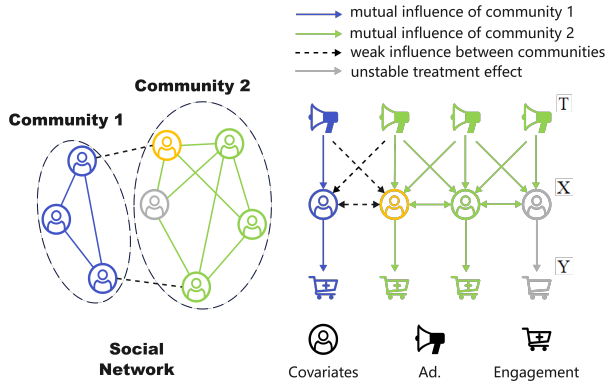

Figure 1: Mutual influence among users in a social network. Users in community 1 (blue) and community 2 (green) exhibit different treatment response patterns. The orange user is the estimation target. Grey shading indicates higher outcome noise (heteroscedasticity).

the same community often amplify treatment effects through social contagion [4, 10, 21].

While Graph Neural Networks (GNNs) have enabled the integration of network structures into causal inference, three major challenges remain. *First*, existing GNN-based ITE estimators treat all neighbors uniformly during message passing, overlooking the hierarchical nature of community-driven interactions. The lack of explicit positional encoding fails to disentangle the influence of high-level community biases from local node interactions [27]. *Second*, real-world outcomes frequently exhibit heteroscedasticity [1]: the behavior patterns of some users may become highly unstable due to measurement errors or anomalous factors, causing significant randomness in treatment outcomes that degrades model performance if left unaddressed. While variance-weighted methods [22, 26] have been explored in traditional ITE estimation, their integration with graph-based approaches remains nascent. *Third*, and perhaps most fundamentally, each new graph ITE model proposes a *monolithic* architecture whose innovations are tightly coupled to the specific design. An improvement that works for one model cannot be trivially transferred to another, limiting cumulative progress in the field.

To address these gaps, we take a different approach from the prevailing paradigm of designing yet another end-to-end architecture. Instead, we propose two orthogonal, *plug-and-play* modules that enhance any existing graph ITE estimator: (1) **Graph Positional Encoding (GPE)**, a self-attention mechanism that fuses node features with node2vec [13] positional embeddings to capture community effects before GNN processing; and (2) **Variance-Weighted Learning**, which estimates per-sample ITE variance and reweights the regression loss accordingly, with theoretical guarantees from

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the Gauss–Markov theorem [2]. Crucially, GPE operates at the *input* level and variance weighting at the *loss* level—neither modifies the base model's internal architecture.

We validate this modular paradigm through a rigorous controlled evaluation across eight base models—spanning the foundational NetDeconf [14] (WSDM 2020) to the current SOTA GDC [15] (WSDM 2025)—on five graph and six tabular datasets, totaling over 1,100 experimental runs. Our key findings are: (i) GPE improves 7 of 8 base models, with TARNet+GPE achieving a *perfect* 11/11 win rate across all graph settings; (ii) Variance weighting provides complementary gains under heteroscedastic noise, yielding a 4× Qini improvement on continuous real-world outcomes; (iii) The combination reduces GDC's PEHE by up to 31% at high noise. Code and full results are available at https://anonymous.4open.science/r/CAVIN.

2 Related Work

ITE estimation and uplift modeling. Traditional approaches encompass tree-based methods [5, 24], knapsack formulations [3, 12], and meta-learners [18, 19]. Deep learning extensions—BNN [17], TARNet/CFRNet [25], SNet [9], TNet [7], and EFIN [20]—primarily adopt the meta-learner framework, utilizing neural networks to estimate the components defined by S-, T-, X-, or DR-learners. However, in domains such as online marketing, samples often violate the i.i.d. assumption, making network interference a critical frontier.

Causal inference under network interference. Early work by Hudgens & Halloran [16] introduced partial interference within predefined clusters. Guo et al. [14] pioneered GCN-based network deconfounding with Wasserstein representation balancing, establishing the BlogCatalog and Flickr benchmarks that subsequent work builds upon. GIAL [8] incorporated mutual-information maximization and adversarial balancing to handle imbalanced network structures. GNUM [30] proposed a transformed-target estimator for direct uplift prediction. IGL [28] introduced environment-invariant representations for inductive cross-domain settings. The current state-of-the-art, GDC [15], disentangles node features into adjustment and confounder representations via learnable masks and treatment-conditioned graph aggregators. A common thread across all these methods is that each proposes a complete, indivisible architecture.

Our positioning. Unlike prior work that introduces a new monolithic model, we contribute two modular enhancements and validate them across the full spectrum of existing architectures, from the 2020 foundational work to the 2025 SOTA. To our knowledge, this is the first controlled "plug-in" evaluation in graph-based ITE estimation.

3 Preliminaries

3.1 Graph-Based ITE Estimation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ represent a graph with node set $\mathcal{V}$ and edge set $\mathcal{E}$. Each node i is characterized by a feature vector $X_i \in \mathbb{R}^d$, a treatment indicator $T_i \in \{0, 1\}$, and an observed outcome Y_i. We define the dataset as $\mathcal{D} = \{(X_i, T_i, Y_i), \mathcal{A}\}_{i=1}^n$, where $\mathcal{A}$ denotes the adjacency matrix. Unlike the Stable Unit Treatment Value Assumption (SUTVA) used in traditional ITE, outcomes in networks are affected by peer effects. We adopt the following assumptions:

[Network Consistency] The potential outcome Y_i depends on the individual's own treatment T_i and an aggregation of their neighbors' treatments: $Y_i = Y_i(T_i, \mathcal{T}_i)$, where $\mathcal{T}_i = \mathrm{agg}(T_{\mathcal{N}_i^{1,\dots,r}})$.

[Network Unconfoundedness] Conditional on individual features X_i and multi-hop neighbor features $X^i_{\mathcal{N}}$, treatment assignment is independent of potential outcomes: $Y_i(T_i, \mathcal{T}_i) \perp (T_i, T^i_{\mathcal{N}}) \mid X_i, X^i_{\mathcal{N}}$.

[Network Overlap] Every treatment combination has non-zero probability: $0 < p(T_i, T^i_{\mathcal{N}} \mid X_i, X^i_{\mathcal{N}}) < 1$.

The objective is to estimate the Network CATE: $\tau(X_i) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i, X^i_{\mathcal{N}}]$. Since the ground truth τ is unobservable, we rely on a proxy variable $\tilde{D}$ (imputed treatment effect) for optimization:

Theorem 3.1 (Optimization Equivalence). *If* $\mathbb{E}[\tilde{D} \mid X] = \tau(X)$, *then* $\arg\min_{\hat{\tau}} \mathbb{E}[(\tilde{D} - \hat{\tau})^2] = \arg\min_{\hat{\tau}} \mathbb{E}[(\tau - \hat{\tau})^2]$.

This justifies using imputed residuals as regression targets in the X-learner framework, which estimates $\mu_t = \mathbb{E}[Y \mid T = t, X, X_{\mathcal{N}}]$ for $t \in \{0, 1\}$, computes cross-residuals $\mathrm{D}_1 = Y(1) - \mu_0$ and $\mathrm{D}_0 = \mu_1 - Y(0)$, and combines them via propensity-weighted regression.

3.2 Graph Attention Networks

To model complex network interference, we adopt multi-head Graph Attention Networks (GATs) [29]. Each node's representation is iteratively updated by aggregating transformed features from its neighborhood:

$$h_v^{(l+1)} = \sigma\left(\sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(l)} W^{(l)} h_u^{(l)}\right), \tag{1}$$

where $\alpha_{vu}^{(l)} = \mathrm{softmax}_u\Big(\mathrm{LeakyReLU}(\mathbf{a}^T [W^{(l)} h_v \| W^{(l)} h_u])\Big)$ are learned attention coefficients. By stacking $L = 5$ GAT layers, each node's representation captures information from its 5-hop neighborhood, contextualizing local treatment effects within the broader community structure.

4 Method

We present two plug-in modules. GPE operates at the *input* level; variance weighting operates at the *loss* level. Neither modifies the internal architecture of any base model.

4.1 Graph Positional Encoding (GPE)

Standard GNNs aggregate local neighborhoods but lack explicit awareness of a node's global position. Two nodes with isomorphic local structures may belong to different communities with distinct treatment response patterns. GPE addresses this by fusing raw features $h \in \mathbb{R}^{n \times d_h}$ with pre-computed node2vec [13] embeddings $p \in \mathbb{R}^{n \times d_p}$ via multi-head cross-attention, where features serve as queries and positional embeddings as keys and values:

$$Q = hW_Q,\ K = pW_K,\ V = pW_V, \tag{2}$$

$$\mathrm{head}_k = \mathrm{softmax}\Big(Q_k K_k^\top / \sqrt{d_k}\Big) V_k, \tag{3}$$

$$z = \mathrm{LN}\big([h;\ W_o\,\mathrm{Concat}(\mathrm{head}_1, \dots, \mathrm{head}_H)] + f_{\mathrm{res}}([h; p])\big), \tag{4}$$

where W_Q, W_K, W_V, W_o are learnable projections, LN denotes layer normalization, $[\cdot;\cdot]$ is concatenation, and f_{res} is a residual projection.

The output z replaces h as the input to any downstream GNN backbone.

Plug-in interface. For any base model M that takes node features X as input, the GPE-enhanced version is simply $M_{+\mathrm{GPE}}(\cdot) = M(\mathrm{GPE}(X, p), \cdot)$. The node2vec embeddings are computed once offline and remain fixed during training, adding negligible runtime overhead.

4.2 Variance-Weighted Learning

Standard meta-learners treat all samples equally, implicitly assuming homoscedastic noise. In practice, user behavior volatility varies significantly—some users are consistent while others are erratic. We model the observed pseudo-treatment effect as $D_i = \tau(X_i) + \varepsilon_i$, where $\mathrm{Var}[\varepsilon_i \mid X_i] = \sigma^2(X_i)$ is feature-dependent.

To estimate $\sigma^2(X_i)$, we use the identity $\sigma^2 = \mathbb{E}[D^2 \mid X] - \tau^2(X)$ and train a variance head to regress on the target $D^2 - \hat{\tau}^2$, using a pilot estimate $\hat{\tau}$ from the outcome heads (with gradients detached to prevent contamination). The sample weights are:

$$w_i = \frac{1}{\max\{\hat{\sigma}^2(X_i), \delta\}}, \quad \delta > 0, \tag{5}$$

and the refined ITE estimator solves $\hat{\tau} = \arg\min_{\hat{\tau}'} \sum_i w_i(\mathrm{D}_i - \hat{\tau}'(X_i))^2$.

Binary outcome adaptation. For binary outcomes (e.g., click/conversion in recommendation), the noise variance is analytically determined by the Bernoulli model: $\sigma_t^2(X_i) = \mu_t(X_i)(1 - \mu_t(X_i))$, requiring no learned variance head.

[On the Linearity Assumption] While our framework uses deep neural networks, the theoretical properties of variance reduction are best understood through local linearization. The final layer of any deep network is a linear regressor on learned features. By Taylor's theorem, $\tau(X)$ can be locally approximated as linear. Under this linearization, the Gauss–Markov theorem guarantees that the WLS estimator achieves $\mathrm{Var}(\hat{\beta}_{\mathrm{WLS}}) \preceq \mathrm{Var}(\hat{\beta}_{\mathrm{OLS}})$ in the Löwner order, i.e., it is the Best Linear Unbiased Estimator [2]. The full proof is provided in the code repository.

4.3 Overall Framework

As shown in Figure 2, our framework enhances any base model M through two orthogonal plug-ins. First, node2vec embeddings are fused with raw features via GPE (Eqs. 2–4). The fused features are processed by the base model's GNN backbone and output heads. Finally, the outcome loss is reweighted by inverse variance estimates. The complete training procedure is formalized in Algorithm 1.

Algorithm 1 Plug-in Training Procedure

Input: Data $\mathcal{D} = \{(X_i, T_i, Y_i)\}_{i=1}^n$, base model M
Parameters: Smoothing δ, epochs K
Compute positional embeddings p via node2vec (offline)
Initialize M's parameters + variance head $\hat{\sigma}^2$
for each epoch $k = 1, \ldots, K$ **do**
  $\tilde{X} \leftarrow \mathrm{GPE}(X, p)$ *// input-level plug-in*
  $\hat{\mu}_t, \hat{e} \leftarrow M(\tilde{X}, \mathcal{A})$ *// base model forward*
  Compute residuals $\mathrm{D}_t = (-1)^t[\hat{\mu}_{1-t}(X) - Y]$
  Update $\hat{\sigma}_t^2$ by regressing on $\mathrm{D}_t^2 - \hat{\tau}_t^2$
  $w_i \leftarrow 1/\max(\hat{\sigma}_t^2(X_i), \delta)$ *// loss-level plug-in*
  Update M via $\nabla \sum_i w_i(\mathrm{D}_i - \hat{\tau}(X_i))^2$
end for

5 Experiments

5.1 Experimental Setup

Graph datasets. We use five graph datasets with semi-synthetic outcomes. Three are generated from real citation networks—CoraFull (19,793 nodes), DBLP (17,716), and PubMed (19,717)—using a community-aware data generation process that extends prior work [14] with Louvain-detected communities [6], multi-hop treatment spillover, and per-node heteroscedastic noise controlled by $\rho \in \{5, 10, 30\}$. Two are standard benchmarks: BlogCatalog (5,196 nodes) and Flickr (7,575), following the generation protocol of Guo et al. [14]. The per-dataset influence coefficients are detailed in Table 1.

Table 1: Semi-synthetic data generation parameters. Shared: $C=5$, $\beta_0=-0.3$, $\beta_1=0.3$, $\gamma=0.6$, hops = 5.

Dataset	κ_1	κ_2	κ_3	κ_4
CoraFull	0.2	0.1	0.03	0.05
DBLP	1.5	0.75	0.2	0.3
PubMed	0.15	0.07	0.025	0.035

Tabular datasets. For real-world validation without graph structure, we evaluate on six tabular uplift benchmarks: Criteo Uplift (1M rows, binary conversion), Hillstrom (42K, binary visit and continuous spend), RetailHero X5 (200K, binary purchase), and Lenta (690K, binary response).

Base models. We test eight GNN-based ITE estimators spanning the full timeline of graph ITE research: BNN [17] (S-learner), TARNet [25] (T-learner), X-learner [19], NetDeconf [14], GIAL [8], GNUM [30], GDC [15], and CAVIN (X-learner + GPE + Var). All share the same backbone: 5-layer GAT with hidden dimensions [256,128,128,128,128], 4-head attention, 4-layer MLP heads (hidden 128). GPE uses 128-dim node2vec embeddings with 4-head attention and embed dimension 256.

Training and evaluation. Adam optimizer with learning rate 10^{-3}, weight decay 10^{-4}, 200 epochs, 60/20/20 train/val/test splits, 3 seeds per graph setting and 5 seeds per tabular setting. We report rooted PEHE (↓), |ATE| (↓), and Qini coefficient (↑).

5.2 GPE as Universal Plug-in

Table 2 presents the headline result: adding GPE to each base model at noise level $\rho = 10$. GPE improves every model except BNN, with the largest gains on NetDeconf (CoraFull PEHE: 7.65→4.64, −39%) and GDC on DBLP (3.19→2.70, −15%).

To assess consistency beyond $\rho = 10$, Table 3 reports the win rate of GPE across all 9 (dataset, ρ) cells on CoraFull/DBLP/PubMed and 2 cells on BlogCatalog/Flickr. TARNet+GPE achieves a *perfect* 11/11 win rate. Six of seven non-trivial models achieve $\geq 7/11$.

Why does BNN not benefit? BNN uses a single outcome model $\mu(X, T)$ that concatenates the treatment indicator with features. Positional information from GPE gets mixed into this concatenated representation without differentially modulating the treatment-specific response functions. In contrast, T-learner-based models

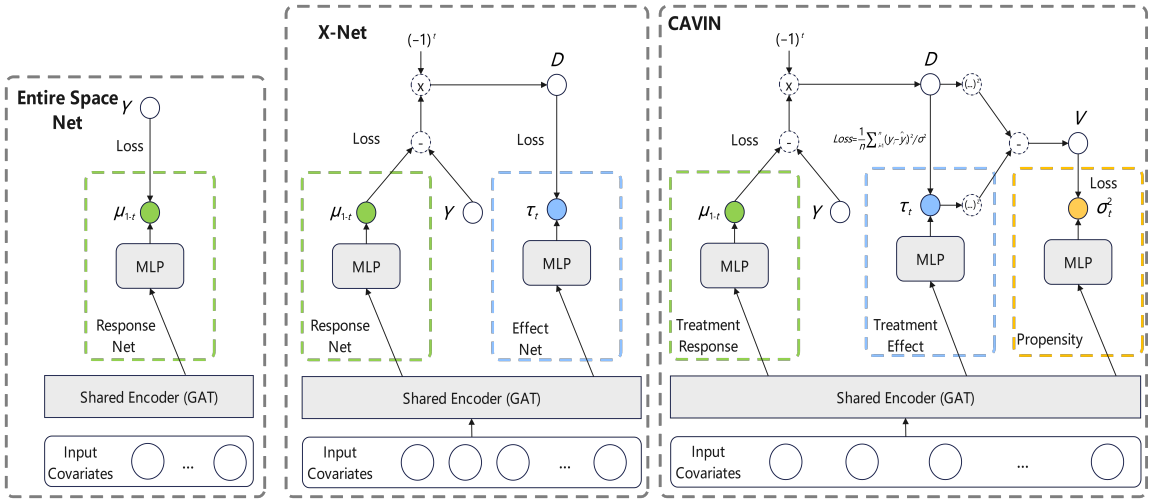

Figure 2: The overall architecture. Left: Comparison with general learners (S/T-Learner) that assume i.i.d. data. Center: The standard X-Learner uses static propensity weights. Right: Our framework adds two plug-in modules—GPE for community-aware feature fusion (input level) and variance-weighted loss for adaptive noise mitigation (loss level)—on top of any base model.

Table 2: Effect of GPE on rooted PEHE (↓) at $\rho = 10$. Format: CoraFull / DBLP / PubMed. Mean over 3 seeds. Bold = GPE improves. ⋆GDC is the current SOTA (WSDM 2025).

	Vanilla PEHE			+GPE PEHE			Avg
Model	CoraFull	DBLP	PubMed	CoraFull	DBLP	PubMed	Δ%
TARNet	3.00	2.52	3.40	**2.64**	**2.27**	**3.23**	−10
GDC⋆	3.91	3.19	3.06	**3.62**	**2.70**	**2.82**	−10
NetDeconf	7.65	3.92	3.33	**4.64**	**3.72**	**3.30**	−21
GIAL	6.18	4.11	3.37	6.27	**3.69**	**3.35**	−5
GNUM	6.09	3.89	4.10	**4.98**	**3.82**	**4.05**	−6
X-learner	3.21	2.31	3.09	**2.62**	**2.25**	3.17	−5
BNN	5.47	5.02	5.45	5.53	5.03	5.45	0

Table 3: GPE win rate: settings where +GPE reduces PEHE.

Model → +GPE	Ours (9)	Blog/Flickr (2)	Total
TARNet	**9/9**	**2/2**	**11/11**
GDC (SOTA)	7/9	1/2	8/11
NetDeconf	7/9	1/2	8/11
GNUM	6/9	2/2	8/11
X-learner	7/9	1/2	8/11
GIAL	7/9	0/2	7/11
BNN (S-learner)	1/9	0/2	1/11

Table 4: GDC ablation at $\rho = 30$ (PEHE ↓). GPE alone overfits on CoraFull/PubMed; adding Var rescues it.

Dataset	GDC	+GPE	+Var	+GPE+Var
CoraFull	4.93	5.90	4.32	**4.21**
DBLP	4.26	3.91	4.30	**3.57**
PubMed	4.31	4.80	3.79	**2.99**

(TARNet, GDC, X-learner) have separate heads for μ_0 and μ_1, allowing each to independently leverage the community context provided by GPE. This finding has practical implications: GPE is most effective when combined with architectures that maintain separate treatment-control representations.

5.3 Variance Weighting: Complementing GPE

While GPE addresses *structural* bias (missing community context), variance weighting addresses *outcome noise*. Table 4 examines their interaction on GDC at high noise ($\rho = 30$), where heteroscedasticity is most pronounced.

A striking pattern emerges: at $\rho = 30$, GPE alone *hurts* GDC on CoraFull (4.93→5.90) and PubMed (4.31→4.80)—the positional attention overfits to noisy labels. Adding variance weighting rescues this: GDC+GPE+Var achieves 2.99 on PubMed, a 31% reduction from vanilla. This demonstrates that the two modules address *orthogonal failure modes*: GPE injects structural information that can amplify noise under extreme heteroscedasticity, while variance weighting dampens exactly this noise. Their combination is synergistic.

Table 5: Tabular variance weighting (Qini ↑, 5 seeds).

Dataset	Outcome	S	S+Var	X	X+Var
Hillstrom	spend	0.029	**0.117**	0.107	0.072
X5	binary	0.013	0.013	0.013	**0.014**
Criteo	binary	**0.167**	0.151	0.151	0.147
Hillstrom	visit	−0.001	−0.005	0.008	**0.009**

Table 6: Full ablation on DBLP $\rho = 30$ (PEHE ↓).

Model	Vanilla	+GPE	+Var	+GPE+Var
TARNet	6.33	5.26	6.51	**4.27**
X-learner	4.66	4.61	**3.84**	4.14
GDC	4.26	3.91	4.30	**3.57**
NetDeconf	4.51	**4.35**	5.04	4.93

5.4 Real-World Tabular Validation

To validate variance weighting without graph structure, we test on real-world tabular benchmarks using meta-learners with and without the Bernoulli variance adaptation ($\sigma^2 = \mu(1-\mu)$). Table 5 reports Qini coefficients.

The headline result is on Hillstrom with continuous spend outcome: S+Var achieves a **4× Qini improvement** (0.029→0.117). This confirms that variance weighting is most effective when the outcome exhibits genuine heteroscedasticity—continuous outcomes with varying noise levels across individuals. For binary outcomes, the improvement is more modest, consistent with the bounded Bernoulli variance. Notably, on the well-balanced Criteo dataset, variance weighting provides no benefit, which is expected: when treatment assignment is already balanced (~85% treated), inverse-variance reweighting has little room to improve.

5.5 Ablation: Component Interactions

Table 6 presents a full ablation on DBLP at $\rho = 30$, the most challenging setting.

Three interaction patterns emerge: (i) For TARNet and GDC, the *combination* outperforms either module alone—they address complementary failure modes. (ii) For X-learner, variance weighting alone achieves the best result—the X-learner's cross-fitting already captures some structural information. (iii) For NetDeconf, GPE alone is optimal—adding variance weighting to this Wasserstein-balanced model disrupts its representation learning. These findings suggest that the optimal plug-in configuration is model-dependent, and practitioners should empirically select based on their base architecture and noise regime.

6 Limitations and Future Work

Our work has three limitations. First, GPE relies on node2vec embeddings that assume a static graph topology; in dynamic networks with evolving connections, embeddings would need periodic recomputation. Second, our evaluation primarily uses semi-synthetic outcomes; deployment-scale validation on a live recommender system remains future work. Third, the variance head introduces additional parameters that may increase overfitting risk on small graphs. Future directions include exploring inductive positional encodings for cold-start settings, integrating GPE with temporal graph networks, and conducting online A/B tests to measure real-world uplift.

7 Conclusion

We introduced Graph Positional Encoding and Variance-Weighted Learning as two orthogonal, architecture-agnostic plug-in modules for graph-based uplift modeling. Through a controlled evaluation on eight base models—spanning five years of graph ITE research—across eleven datasets and over 1,100 runs, we demonstrated that GPE consistently improves treatment effect estimation (7/8 models, up to 39% PEHE reduction), that variance weighting provides complementary gains under heteroscedasticity (4× Qini on real-world continuous outcomes), and that their combination achieves the best results at high noise (up to 31% PEHE reduction on the current SOTA). Rather than proposing another monolithic model, our work advocates for a modular enhancement paradigm that enables cumulative, transferable progress in graph-based causal inference for recommender systems.

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